

HDOC Day 25 - C Programming Activity

Increasing Patterns Using Switch Statement and Nested Loops

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Problem Statement

Write a C program that displays a menu and uses a switch statement to print the selected increasing pattern. Use nested loops without arrays. Each pattern is implemented in a separate parameterless function with no return value.

Patterns: (1) Increasing right-angled triangle, (2) Increasing row-number triangle, (3) Increasing natural-number pattern, (4) Consecutive-number triangle, and (5) Increasing alphabet pattern. For the alphabet pattern, $1 \leq n \leq 26$.

1. Increasing Right-Angled Triangle

Algorithm

1. Start.
2. Input number of rows n .
3. For i from 1 to n , run an inner loop j from 1 to i .
4. Print '*' in each inner-loop iteration.
5. After each row, print a newline.
6. Stop.

Test-Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	$n = 1$	*	*	Pass
2	$n = 4$	1, 2, 3, 4 stars by row	1, 2, 3, 4 stars by row	Pass
3	$n = 5$	5 rows increasing stars	5 rows increasing stars	Pass

2. Increasing Row-Number Triangle

Algorithm

1. Start.
2. Input number of rows n .
3. For i from 1 to n , run j from 1 to i .
4. Print the value of i in the inner loop.
5. Print a newline after each row.
6. Stop.

Test-Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	$n = 1$	1	1	Pass
2	$n = 4$	Rows: 1; 2 2; 3 3 3; 4 4 4 4	Same as expected	Pass
3	$n = 5$	5 rows; row value repeated row times	Same as expected	Pass

3. Increasing Natural-Number Pattern

Algorithm

1. Start.
2. Input number of rows n.
3. For i from 1 to n, run j from 1 to i.
4. Print j in each inner-loop iteration.
5. Print a newline after each row.
6. Stop.

Test-Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n = 1	1	1	Pass
2	n = 4	Rows end at 1, 2, 3, 4	Rows end at 1, 2, 3, 4	Pass
3	n = 5	Each row prints 1 through row number	Same as expected	Pass

4. Consecutive-Number Triangle

Algorithm

1. Start.
2. Input number of rows n and initialize num = 1.
3. For i from 1 to n, run j from 1 to i.
4. Print num, then increment num by 1.
5. Print a newline after each row.
6. Stop.

Test-Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n = 1	1	1	Pass
2	n = 4	1 / 2 3 / 4 5 6 / 7 8 9 10	Same as expected	Pass
3	n = 5	Consecutive numbers through 15	Consecutive numbers through 15	Pass

5. Increasing Alphabet Pattern

Algorithm

1. Start.
2. Input number of rows n.
3. If n is outside 1 to 26, print an invalid-input message and return.
4. For i from 1 to n, run j from 1 to i.
5. Print character 'A' + j - 1.
6. Print a newline after each row.
7. Stop.

Test-Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n = 1	A	A	Pass
2	n = 4	A / A B / A B C / A B C D	Same as expected	Pass
3	n = 27	Invalid number of rows	Invalid number of rows	Pass

Complete C Program

```
#include <stdio.h>

void pattern1()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

void pattern2()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++)
            printf("%d ", i);
        printf("\n");
    }
}

void pattern3()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

void pattern4()
{
    int n, i, j, num = 1;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) {
            printf("%d ", num);
            num++;
        }
        printf("\n");
    }
}

void pattern5()
{
    int n, i, j;
    printf("Enter number of rows (1-26): ");
    scanf("%d", &n);
    if (n < 1 || n > 26) {
        printf("Invalid number of rows.\n");
        return;
    }
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++)
            printf("%c ", 'A' + j - 1);
        printf("\n");
    }
}

int main()
{
    int choice;
    printf("INCREASING PATTERNS MENU\n");
    printf("1. Increasing right-angled triangle\n");
    printf("2. Increasing row-number triangle\n");
    printf("3. Increasing natural-number pattern\n");
    printf("4. Consecutive-number triangle\n");
    printf("5. Increasing alphabet pattern\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
```

```
switch (choice) {
    case 1: pattern1(); break;
    case 2: pattern2(); break;
    case 3: pattern3(); break;
    case 4: pattern4(); break;
    case 5: pattern5(); break;
    default: printf("Invalid choice.\n");
}
return 0;
}
```

Program Execution

```
Enter your choice: 1
Enter number of rows: 4
* * * *
* * *
* *
*
(dhruvpandey@DHRUVs-MacBook-Air ~ % ./day25
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4
(dhruvpandey@DHRUVs-MacBook-Air ~ % ./day25
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 4
1
1 2
1 2 3
1 2 3 4
(dhruvpandey@DHRUVs-MacBook-Air ~ % ./day25
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 4
1
2 3
4 5 6
7 8 9 10
(dhruvpandey@DHRUVs-MacBook-Air ~ % ./day25
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter number of rows (1-26): 5
A
A B
A B C
A B C D
A B C D E
dhruvpandey@DHRUVs-MacBook-Air ~ %
```